Sakshi D. Maske

Independent Geospatial Researcher

Abstract

Small island nations are widely assumed to face a compounding climate risk: high physical exposure to sea-level rise layered with degrading natural coastal defenses. This assumption is rarely tested against independent, multi-temporal evidence, and rarely disaggregated by ecosystem type. This study tests the compounding-vulnerability hypothesis across five island nations spanning three ocean basins — Maldives, Lakshadweep, Seychelles, Fiji, and the Canary Islands — treating mangrove and coral reef degradation as two independent pathways rather than a single undifferentiated category. Physical exposure was quantified from settlement-level elevation data; mangrove extent was tracked across three independent time points (1996, 2010, 2020); and coral condition was tracked using a continuous 24-year satellite-derived thermal-stress record. Coral reef systems show a measurable, rising bleaching-stress trend in four of five islands, most severely in Seychelles. Mangrove extent, tested with equal rigor, shows no measurable decline in any of the three islands where mangroves are present — a result that does not support the assumption of uniform ecosystem decline. Combining physical exposure with coral degradation into a composite vulnerability score reveals that the island with the highest physical exposure (Maldives) is not the island with the highest overall compound risk (Seychelles), demonstrating that exposure alone is an incomplete vulnerability measure. A supplementary test of whether formal protected-area coverage aligns with this empirically verified risk finds a positive but not statistically significant relationship, consistent with a small-sample limitation rather than a genuine absence of governance responsiveness. These findings indicate that coastal ecosystem degradation is not a uniform phenomenon across ecosystem types, with direct implications for how adaptation and conservation resources should be prioritized.

Keywords: sea-level rise, mangroves, coral bleaching, compound vulnerability, small island states, protected-area effectiveness, remote sensing

---

1. Introduction

International climate adaptation policy increasingly treats coastal ecosystems — mangroves, coral reefs, seagrass beds — as a single protective category, financed and managed under the umbrella term “nature-based solutions.” A substantial body of research documents that these ecosystems genuinely do provide measurable coastal protection: reef structures dissipate wave energy before it reaches shore, and dense mangrove stands can reduce wave heights by well over half and, where wide enough, storm-surge peak water levels as well. Small island developing states are frequently identified as the setting where this protective value is proportionally greatest relative to national economic output.

What is less frequently tested is whether the ecosystems providing this protection are themselves degrading uniformly, or whether adaptation planning that treats them as a single undifferentiated buffer risks obscuring materially different underlying trajectories. This study addresses that gap directly, testing mangrove extent and coral reef condition as two independent pathways rather than assuming they move together, across a genuinely cross-national, multi-ocean-basin sample.

2. Literature Review

2.1 Nature-Based Coastal Protection and Its Assumed Uniformity

Coastal ecosystems are established in the literature as effective, measurable protective infrastructure. Reef structures have been shown to reduce incoming wave energy substantially, while mangrove belts of sufficient width can reduce wave heights and storm-surge peak levels meaningfully per kilometer of forest. Comparative work assessing corals, seagrasses, and mangroves together finds that combining multiple habitat types delivers more protection than any single habitat alone, and specifically cautions against the common practice of evaluating nature-based protection through a single-habitat lens rather than a whole-system approach — a caution directly relevant to this study's decision to test mangrove and coral trajectories independently rather than as one combined signal.

2.2 Global Mangrove Trends: A More Stable Picture Than Often Assumed

While mangrove loss driven by aquaculture and agricultural conversion is well documented across the twentieth century, the most comprehensive recent multi-decadal remote-sensing assessment — the Global Mangrove Watch archive's own change-detection analysis spanning 1996 to 2020 — found a global net loss of only approximately 3.4%, a considerably more modest figure than earlier deforestation-era estimates might suggest, indicating that mangrove loss has slowed substantially in the more recent multi-decade record relative to the mid-twentieth-century period. This finding is directly relevant to the present study's own result of no measurable mangrove decline across the three tested islands: rather than being an anomalous or surprising finding, it is broadly consistent with the direction, if not necessarily the exact magnitude, of the most authoritative recent global assessment of the same dataset this study draws from.

2.3 Coral Bleaching as a Distinct, Escalating Pathway

In contrast to the comparatively stable recent mangrove picture, coral bleaching driven by thermal stress is documented as increasing in both frequency and severity across recent decades, a trend attributed directly to rising ocean temperatures. NOAA Coral Reef Watch's satellite-derived Degree Heating Week product, the same metric used in this study, is established in the literature as a validated predictor of bleaching intensity, with values above 4°C-weeks associated with significant bleaching risk and values above 8°C-weeks associated with reef-wide bleaching and heat-sensitive coral mortality. The global coral reef system is, at the time of this study, in the midst of a fourth documented global bleaching event beginning in 2023, with bleaching confirmed across dozens of countries and all three major ocean basins — providing important real-world context for this study’s finding of a rising thermal-stress trend in four of the five tested islands.

2.4 Protected-Area Effectiveness and the “Paper Park” Problem

A separate, substantial body of literature documents that formal protected-area designation frequently fails to translate into effective on-the-ground management, a phenomenon widely termed the “paper park” problem — protected areas that exist in legal or administrative terms without correspondingly effective enforcement, monitoring, or ecological outcomes. Recent global assessments estimate that while roughly ten percent of the ocean carries some form of formal protected status, only a small fraction of that area meets rigorous effectiveness standards. Identified drivers of this implementation gap include inadequate enforcement capacity, insufficient stakeholder engagement, and a persistent disconnect between protected-area evaluation findings and actual management action. This literature motivates the present study’s governance-alignment test: rather than treating formal protection status as inherently meaningful, this study tests whether protected-area coverage is empirically aligned with independently verified ecological and physical risk, directly in the spirit of the paper-park critique.

3. Data and Methodology

3.1 Study Design

This study tests compound climate vulnerability across five island nations — Maldives, Lakshadweep, Seychelles, Fiji, and the Canary Islands — selected for their span across three distinct ocean basins and their geological diversity, ranging from low-lying coral atolls to volcanic, mountainous terrain. Two ecosystem pathways (mangrove extent, coral thermal condition) are tested independently rather than combined into a single assumed-uniform “ecosystem buffer” variable.

3.2 Data Sources

Variable	Source	Temporal Coverage
Settlements, elevation	OpenStreetMap; Copernicus DEM GLO-30	Current
Mangrove extent	Global Mangrove Watch	1996, 2010, 2020
Coral thermal stress	NOAA Coral Reef Watch (Degree Heating Week)	1996–2020, continuous
Protected areas	World Database on Protected Areas	Current
Settlement/infrastructure change	Sentinel-2 (NDBI)	2016 vs. 2024

3.3 Physical Exposure

Elevation was sampled at every settlement location across all five islands, and the proportion of settlements at or below a standard one-meter sea-level-rise threshold was calculated per island.

As a complementary metric, physical exposure was also computed on a population-weighted basis: WorldPop 2020 population raster cells were classified as at-risk where their corresponding elevation (resampled to the population grid) fell at or below the one-meter threshold, and the proportion of total population at risk was calculated directly from population counts rather than settlement counts. This was computed for all five islands; Lakshadweep's population raster — previously unavailable due to the impractical file size of India's national dataset — was obtained via a subset clipped from a smaller regional file. For Fiji, elevation coverage did not extend to the easternmost Lau Islands (beyond the antimeridian); Fiji's population-weighted exposure therefore reflects approximately 97.6% of its national population, with the excluded portion reported explicitly.

3.4 Mangrove Extent

Rather than relying on a single before/after comparison, mangrove area was independently measured at three time points (1996, 2010, 2020) in an equal-area projection, avoiding the distortion introduced by measuring change through raw polygon counts, which can vary with satellite classification segmentation behavior independent of any genuine change in underlying area.

3.5 Coral Thermal Stress

Coral condition was measured using the continuous Degree Heating Week time series rather than mapped physical extent, consistent with the understanding that coral degradation manifests primarily as thermally driven bleaching rather than area loss. An early reference period (1996–2000) was compared against a recent period (2016–2020) for each island.

As a robustness check, a Mann-Kendall trend test (non-parametric, standard for environmental time-series analysis) was applied to the complete 24-year DHW series for each island, in addition to the reference-period comparison reported above.

3.6 Compound Vulnerability Score

Physical exposure and coral thermal-stress trend were normalized to a common 0–1 scale using min-max normalization and combined with equal weighting into a single Compound Vulnerability Score per island. Mangrove trend was not included as a weighted input, given the absence of any measurable decline to weight. To test sensitivity to this equal-weighting choice, the composite ranking was recomputed across the full 0–100% weighting range between the two input variables. Seychelles remained the highest-ranked island for the large majority of that range — from 0% up to approximately 76.8% physical-exposure weighting — with Maldives overtaking it only beyond that point, i.e. only if physical exposure were weighted at roughly three-quarters or more of the composite score. At the equal 50/50 weighting actually used in this study, Seychelles is unambiguously highest-ranked, and the crossover point is far enough from 50/50 that the central finding is not an artifact of the specific weighting chosen — but it is not accurate to say Seychelles leads across the entire range. Equal weighting was adopted as a conservative baseline, since no established literature provides a robust empirical basis for differential weighting between physical exposure and coral thermal stress in this specific cross-national context; the sensitivity analysis in Section 4.6 gives the exact range over which this choice holds.

The one-meter sea-level-rise threshold used throughout this study is itself a modeling choice rather than an exact prediction, so settlement-level exposure was also recomputed at 0.5m and 1.5m thresholds, spanning the practical range of near-term projections. Section 4.6 reports this check in full.

3.7 Governance Alignment

Protected-area coverage was quantified within a ten-kilometer coastal buffer around each island (rather than total captured extent, which for some islands included large offshore marine zones only loosely relevant to settlement-level risk) and tested against the Compound Vulnerability Score using a Pearson correlation.

4. Results

4.1 Physical Exposure

Settlement-level exposure to a one-meter sea-level-rise threshold ranged widely: 99.1% (Maldives), 78.3% (Seychelles), 77.8% (Lakshadweep), 32.0% (Fiji), and 0.3% (Canary Islands) — directly reflecting the underlying geological distinction between low-lying coral atoll nations and volcanic, mountainous terrain. (A small cluster of literal-zero-elevation DEM readings at Canary Islands settlement points was excluded as a verified NoData artifact rather than genuine near-sea-level terrain — Canary Islands has no coastline low enough to plausibly produce this many settlements at exactly 0m on a mountainous, volcanic island; see Section 6.)

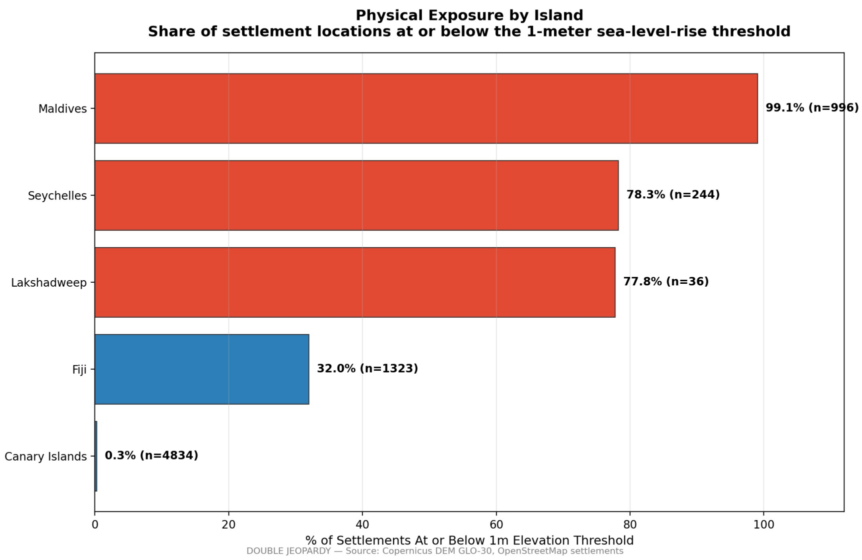

Figure 1. Physical exposure by island, measured as the share of settlement locations at or below the 1-meter sea-level-rise threshold. Low-lying coral atoll nations (Maldives, Seychelles, Lakshadweep) show substantially higher exposure than volcanic, mountainous islands (Fiji, Canary Islands).

Island	Settlement-based exposure	Population-weighted exposure

Maldives	99.1%	64.5%
Seychelles	78.3%	17.6%
Lakshadweep	77.8%	87.5%
Fiji	32.0%	2.1%*
Canary Islands	0.3%	1.6%

*covers ~97.6% of Fiji's population; see Limitations.

Notably, the population-weighted ranking diverges materially from the settlement-based ranking: Lakshadweep, third by settlement-based exposure, becomes the highest population-weighted exposure island, while Fiji and the Canary Islands show even lower relative exposure once weighted by population. This indicates that where people are concentrated within an island's settlement pattern matters independently of how many settlement locations fall below the threshold — reinforcing this study's broader finding that single-indicator exposure measures can misrepresent true risk.

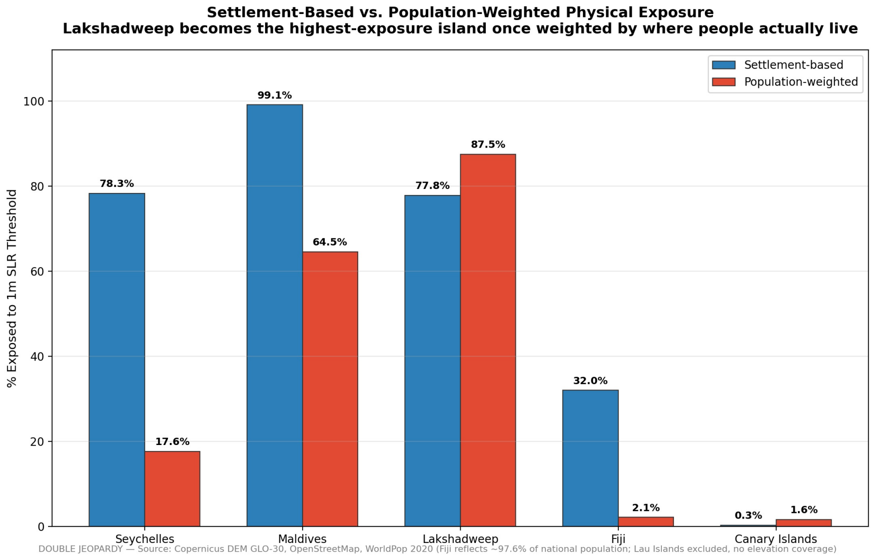

Figure 2. Settlement-based versus population-weighted physical exposure by island. Lakshadweep becomes the highest-exposure island once weighted by where population is actually concentrated, while the Maldives' exposure drops from 99.1% to 64.5% — demonstrating that settlement-count exposure alone can misrepresent the population actually at risk.

4.2 Mangrove Extent: No Measurable Decline

Across all three independent time points and all three islands with mangroves present, area remained essentially stable: Maldives (0.97 km^2 at all three points), Seychelles (3.83–3.84 km^2), and Fiji (485.7 km^2 in 1996 to 488.4 km^2 in 2020, a net increase of 0.6%). This finding does not support the hypothesis that mangrove ecosystems in this sample are measurably declining.

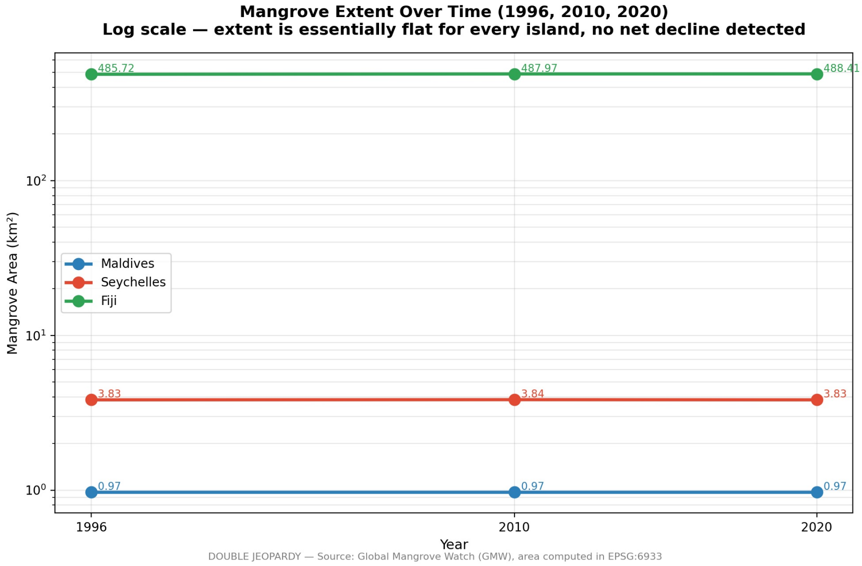

Figure 3. Mangrove extent (km^2, log scale) at three independent time points (1996, 2010, 2020) for the three islands where mangroves are present. Extent remains essentially flat for all three islands, with no measurable decline detected.

4.3 Coral Thermal Stress: A Rising Trend

Comparing early-period and recent-period averages, four of five islands showed increasing thermal stress: Maldives (+0.17°C-weeks), Fiji (+0.10), Lakshadweep (+0.08), and Seychelles (+0.68, with a maximum single recorded value of 10.47°C-weeks — within the range associated with severe bleaching and multi-species mortality). The Canary Islands showed a slight decline (−0.05), consistent with its distinct Atlantic climate regime relative to the four Indian Ocean and Pacific islands.

When tested using a Mann-Kendall trend test on the complete 24-year time series, the increasing trend was statistically significant for two islands — Maldives ($p=0.011$) and Seychelles ($p=0.025$), the two islands central to this study's compound vulnerability ranking — while the more modest increases observed in Fiji and Lakshadweep did not reach statistical significance over the full series ($p=0.184$ and $p=0.386$, respectively).The Canary Islands showed no significant trend ($p=0.641$; Sen's slope ≈ 0), in contrast to the slight decline suggested by the simpler period-comparison method above — indicating that Canary Islands' coral thermal stress has no reliable directional trend over the full 24-year record

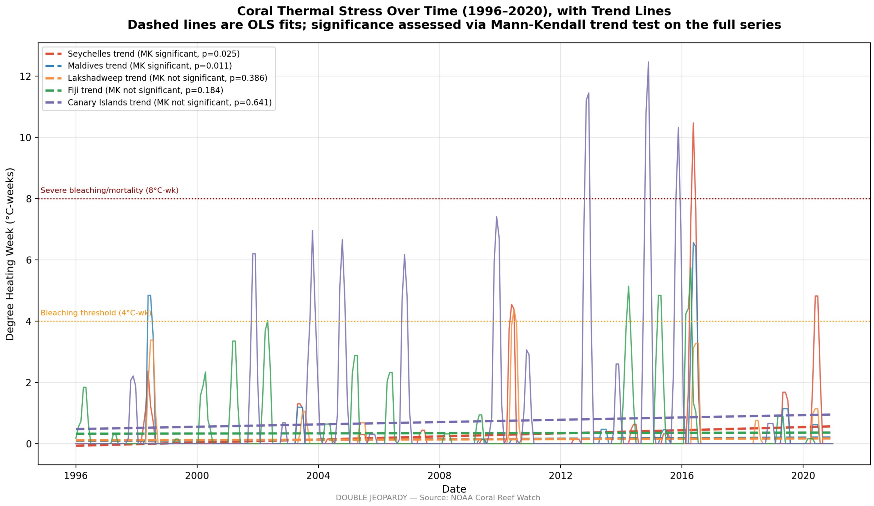

Figure 4. Coral thermal stress (Degree Heating Week) over the full 1996–2020 record for all five islands, with OLS trend lines. The Mann-Kendall trend test finds a statistically significant increasing trend for Maldives (p=0.011) and Seychelles (p=0.025) only.

4.4 Compound Vulnerability: Exposure Alone Is Insufficient

The composite score ranked Seychelles highest (0.895), followed by Maldives (0.651), Lakshadweep (0.481), Fiji (0.263), and Canary Islands (0.000). This directly demonstrates that physical exposure alone would misidentify the highest-risk island: Maldives has substantially higher exposure alone (99.1% versus 78.3%), yet Seychelles produces the higher composite score once its more severe coral degradation trend is incorporated.

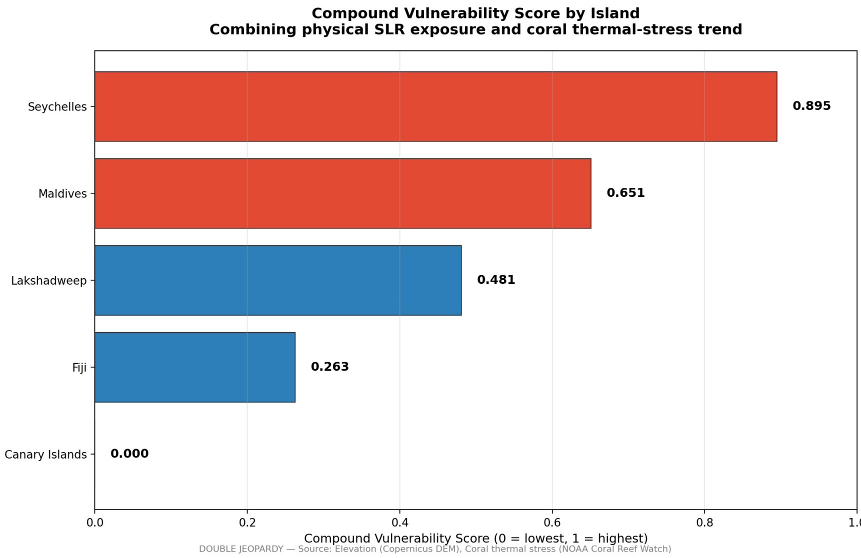

Figure 5. Compound Vulnerability Score across the five study islands, combining normalized physical sea-level-rise exposure and long-term coral thermal-stress trend using equal weighting. Seychelles emerges as the most vulnerable island despite Maldives having the highest physical exposure alone, demonstrating that exposure by itself is an incomplete measure of climate vulnerability and reinforcing the need for a multi-indicator assessment framework.

4.5 Governance Alignment: Suggestive, Not Confirmatory

The correlation between compound vulnerability and coastal protected-area coverage was moderately positive (r=0.718) but did not reach conventional statistical significance (p=0.172), a result attributable to the study's necessarily small five-island sample rather than to an absence of any underlying relationship.

The 95% confidence interval for this correlation, computed via Fisher's z-transformation, spans from r = -0.45 to r = 0.98 — illustrating that with only five data points, the point estimate of r = 0.718 carries very little precision, and the true underlying relationship could plausibly range from weakly negative to nearly perfect positive.

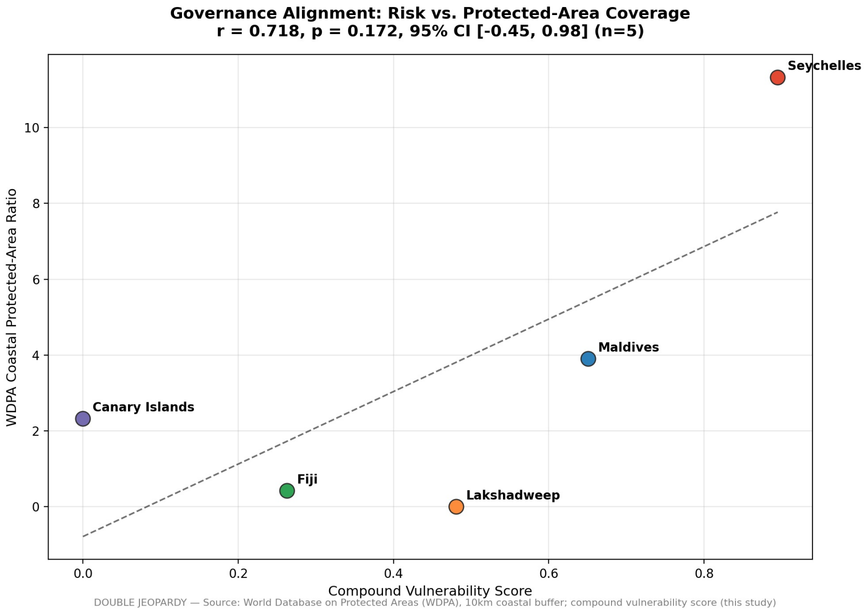

Figure 6. Compound Vulnerability Score plotted against the coastal WDPA protected-area ratio for each island, with a fitted OLS reference line. r = 0.718, p = 0.172, 95% CI [-0.45, 0.98] (n=5) — a positive but statistically inconclusive relationship, driven by the small sample size rather than a null result.

4.6 Robustness and Sensitivity Checks

Several additional checks were run to test the sensitivity of this study's central findings to specific methodological choices:

— Compound Vulnerability Score weighting (Section 4.4): recomputed across the full 0–100% weighting range between physical exposure and coral thermal stress; Seychelles remained the highest-ranked island from 0% up to ~76.8% physical-exposure weighting, with Maldives overtaking it only beyond that point (a later verification pass corrected this from an earlier, less precise claim that Seychelles led across the entire range — see DJ_Development_Log.md).

— Coral thermal-stress trend (Section 4.3): tested using a Mann-Kendall trend test on the complete 24-year time series in addition to the period-comparison method; the increasing trend reached statistical significance for Maldives (p=0.011) and Seychelles (p=0.025) — the two islands central to the compound vulnerability ranking.

— Physical exposure measurement (Section 4.1): recomputed on a population-weighted basis in addition to the settlement-count basis, using WorldPop 2020 data for all five islands.

— Physical exposure threshold (Section 4.1): the 1-meter sea-level-rise threshold used throughout this study was recomputed at 0.5m and 1.5m to test whether the specific threshold choice was doing hidden work in the result. Across this range, each island's exposure percentage moves by at most half a percentage point, and the island ranking never changes at any of the three thresholds tested:

Island	0.5m	1.0m (used in this study)	1.5m
Maldives	99.0%	99.1%	99.1%
Seychelles	78.3%	78.3%	78.3%
Lakshadweep	77.8%	77.8%	77.8%
Fiji	31.4%	32.0%	32.0%
Canary Islands	0.1%	0.3%	0.3%

— Governance-alignment correlation (Section 4.5): the 95% confidence interval (Fisher's z-transformation) for r=0.718 was computed explicitly ([-0.45, 0.98]), making the small-sample limitation quantitatively concrete rather than only qualitatively noted.

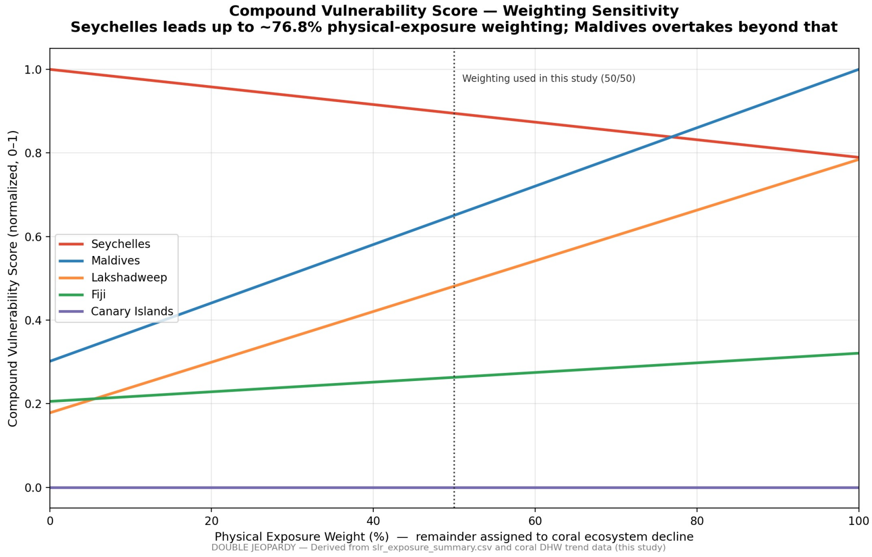

Figure 7. Compound Vulnerability Score for each island as the weighting between physical exposure and coral thermal stress is swept continuously from 0% to 100%. Seychelles remains the highest-ranked island across roughly the first three-quarters of the range (up to ~76.8% physical-exposure weighting); Maldives overtakes it only beyond that point. Since the crossover point is well away from the 50/50 weighting actually used in this study, the central finding is not an artifact of the specific weighting chosen.

Together, these checks indicate that the study's central finding — that Seychelles carries the highest compound vulnerability despite lower physical exposure than the Maldives — is robust to the specific weighting choice, insensitive to the exact sea-level-rise threshold used, and supported by a statistically significant trend test, while more exploratory components of the analysis (governance alignment, the smaller coral trends in Fiji and Lakshadweep) are appropriately reported with their associated uncertainty rather than overstated.

5. Discussion

The central finding — that ecosystem degradation is not uniform across type — is directly consistent with the divergent literature trajectories reviewed above: recent global mangrove assessments describe a comparatively modest net loss over the same multi-decade period this study examines, while coral bleaching literature describes an escalating, currently ongoing global event. The present study's island-level findings track these global patterns closely rather than diverging from them, lending external credibility to the results.

The governance-alignment finding, while not statistically significant, is directionally consistent with a growing literature documenting that protected-area coverage frequently fails to track genuine ecological risk — the "paper park" phenomenon. That Seychelles, the highest-vulnerability island in this sample, also carries the highest coastal protection ratio is a modestly encouraging signal against that broader pattern, though the small sample size means this cannot be treated as confirmed evidence of risk-responsive governance.

6. Limitations

The governance-alignment test is limited by a necessarily small five-island sample, providing insufficient statistical power to confirm a relationship that is nonetheless directionally positive. Population-weighted exposure for Fiji reflects approximately 97.6% of the island's population; elevation data did not cover Fiji's easternmost territory (the Lau Islands, beyond the antimeridian), and this excluded population is reported explicitly rather than assumed negligible. This gap does not affect any other analysis in this study, which relies on settlement point locations rather than the population raster.

Canary Islands' settlement-based exposure figure required a specific data-quality correction: 649 of 5,483 settlement points returned a literal 0m elevation reading from the DEM, which — given Canary Islands' volcanic, mountainous terrain, where genuine sea-level settlements are not expected — was verified as a NoData artifact rather than real terrain, and excluded from the exposure calculation. This does not affect the study's central finding, since Canary Islands was already the lowest-exposure, lowest-vulnerability island in the sample before this correction, but it does move its reported exposure figure from an earlier, artifact-inflated 12.1% down to the corrected 0.3%.

Physical exposure was estimated using Copernicus DEM GLO-30, a 30-meter-resolution, radar-derived global elevation model that captures surface elevation — including vegetation canopy and built structures — rather than true bare-earth elevation. This introduces non-trivial vertical uncertainty relative to the fine, one-meter threshold used to classify settlement exposure, a limitation well documented in prior assessments of global elevation models applied to low-elevation coastal zones. Accordingly, the reported exposure percentages should be interpreted as directionally reliable — supporting the relative ranking of islands that drives this study's central finding — rather than as precise absolute counts.

7. Conclusion

This study finds that small island nations do face a compounding vulnerability to climate change, but that this compounding is neither uniform across ecosystem type nor adequately captured by physical exposure alone. Coral reef degradation is measurable and increasing across most of the sample; mangrove extent, tested with equal rigor across multiple independent time points, shows no comparable decline. The resulting implication for adaptation and conservation policy is direct: resources allocated under an assumption of uniform ecosystem risk may be systematically misallocated relative to where degradation is actually occurring.

References

Bunting, P., Rosenqvist, A., Hilarides, L., Lucas, R. M., Thomas, N., Tadono, T., Worthington, T. A., Spalding, M., Murray, N. J., & Rebelo, L. M. (2022). Global Mangrove Extent Change 1996–2020: Global Mangrove Watch Version 3.0. Remote Sensing, 14(15), 3657. https://doi.org/10.3390/rs14153657

Ferrario, F., Beck, M. W., Storlazzi, C. D., Micheli, F., Shepard, C. C., & Airoldi, L. (2014). The effectiveness of coral reefs for coastal hazard risk reduction and adaptation. Nature Communications, 5, 3794. https://doi.org/10.1038/ncomms4794

Guannel, G., Arkema, K., Ruggiero, P., & Verutes, G. (2016). The Power of Three: Coral Reefs, Seagrasses and Mangroves Protect Coastal Regions and Increase Their Resilience. *PLOS ONE*, 11(7), e0158094. https://doi.org/10.1371/journal.pone.0158094

Heron, S. F., Maynard, J. A., van Hooidonk, R., & Eakin, C. M. (2016). Warming Trends and Bleaching Stress of the World’s Coral Reefs 1985–2012. *Scientific Reports*, 6, 38402. https://doi.org/10.1038/srep38402

National Oceanic and Atmospheric Administration. (2024, April 15). NOAA Confirms 4th Global Coral Bleaching Event. Read

Pieraccini, M., Coppa, S., & De Lucia, G. A. (2017). Beyond marine paper parks? Regulation theory to assess and address environmental non-compliance. Aquatic Conservation: Marine and Freshwater Ecosystems, 27(1), 177–196. https://doi.org/10.1002/aqc.2632

Pike, B. (2026). 10% Protected. 3% Effective. The Widening Gap We Can’t Ignore. Marine Conservation Institute. Read

The Nature Conservancy, Mapping Ocean Wealth. Coastal Protection: The Role of Mangroves and Coral Reefs. Read